a
1 Ground truth
Validated non-isothermal column model (companion study)
Rate law: LDF (L1) or loading-dependent k(q) (L2)
Langmuir
step
q*/q_max
c/c_in
2 Design grid
Damköhler number Da: 0.1 to 1000, nine levels
Péclet number: about 470, 94, 19
Noise σ and isotherm error ε: 0, 0.5, 2 and 5 % each
60 replicates (15 select, 45 estimate)
3 Observation models
O1 probes of c, q, T
O2 probes of c, T (q rebuilt)
O3 outlet c(t), T(t) only
O3
methods get a measured isotherm
4 Can the rate law be discovered?
Sparse regression, weak and strong forms: STLSQ (M2), ensemble (M3), exact best subset (M4)
Errors-in-variables best subset (M5, oracle); PySR (M7); neural network in the column model (M1, demonstration); slow manifold (M9); SINDy-PI (M6, isothermal control)
Success: exactly the true terms, signs, and q* : q = -1 ± 10 %
5 Is the rate constant identifiable?
The true law is given; only k is fitted, on the same observations with the same measured isotherm
Profile likelihood of k (O1 exact; O2 and O3 by full column solves)
Identifiable: 95 % interval inside [k/2, 2k]
6 Boundaries and the analyses that use them (Results subsection)
H2b discovery against identifiability boundary, 96 slices (1) confirmatory
H2a does one error ratio place the boundary? (2) confirmatory
H2c does errors-in-variables fitting move it? (2) confirmatory
which methods discover the law, and where (3) descriptive
O2 and O3: probes without uptake, outlet only (4) descriptive
H2d loading-dependent rate, step isotherm (5) exploratory
b
illustrative curves
k identifiable
Da*ident
Da*disc
discovery of the law
gap
probability
1.0
0.5
0.0
−1
0
1
2
3
log10 Da